

39.4 Bayesian generalized linear models and MCMC

Proposition 39.3.1 closed off the closed form. What is left is a posterior known only up to a constant, and two ways to work with it: approximate it by a normal density at its mode, accurate to order n^{-1} and the origin of the BIC; or sample from it, accurate to whatever the Monte Carlo error allows. This section does both, with the samplers written out rather than delegated.

39.4.1 The posterior

Definition 39.4.1. The Bayesian generalized linear model

Let $\mathbf{y}$ follow the generalized linear model of Definition 34.2.1 with link g , dispersion ϕ and log-likelihood $\ell(\boldsymbol{\beta})$, and let π be a prior density on $\mathbb{R}^p$. The **posterior density** of $\boldsymbol{\beta}$ is

$$\pi(\boldsymbol{\beta} \mid \mathbf{y}) = \frac{\exp\{\ell(\boldsymbol{\beta})\} \pi(\boldsymbol{\beta})}{m(\mathbf{y})}, \quad m(\mathbf{y}) = \int_{\mathbb{R}^p} \exp\{\ell(\mathbf{b})\} \pi(\mathbf{b}) d\mathbf{b}, \quad (39.4.1)$$

whenever $0 < m(\mathbf{y}) < \infty$; the posterior is then **proper**. The **maximum a posteriori** estimate $\hat{\boldsymbol{\beta}}_{\text{MAP}}$ maximizes Eq. 39.4.1, equivalently maximizes the penalized log-likelihood $\ell(\boldsymbol{\beta}) + \log \pi(\boldsymbol{\beta})$. The default choice throughout this chapter is the independent normal prior $\boldsymbol{\beta} \sim N_p(\mathbf{m}, \boldsymbol{\Sigma})$, for which

$$\log \pi(\boldsymbol{\beta} \mid \mathbf{y}) = \ell(\boldsymbol{\beta}) - \frac{1}{2}(\boldsymbol{\beta} - \mathbf{m})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\beta} - \mathbf{m}) + \text{const.} \quad (39.4.2)$$

Because Eq. 39.4.2 is concave whenever ℓ is (Theorem 34.3.3), $\hat{\boldsymbol{\beta}}_{\text{MAP}}$ is found by the Fisher scoring of Theorem 34.4.1 with the score replaced by $\mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu})/\phi - \boldsymbol{\Sigma}^{-1}(\boldsymbol{\beta} - \mathbf{m})$ and the information by $\mathbf{X}^\top \mathbf{W} \mathbf{X}/\phi + \boldsymbol{\Sigma}^{-1}$: one extra term in each. What the prior buys is not only shrinkage but existence.

Proposition 39.4.1. When a flat prior gives a posterior

Let π be a prior on $\mathbb{R}^p$.

- (a) If π is proper and the likelihood is bounded, the posterior is proper. Both the logistic and the Poisson likelihoods are bounded.
- (b) Take the improper prior $\pi \propto 1$ and the ungrouped logistic model with $\text{rank}(\mathbf{X}) = p$. Then the posterior is proper if and only if the data are in overlap in the sense of Definition 35.3.1, that is if and only if the maximum likelihood estimate exists (Theorem 35.3.1).

Proof.

- (a) If $L \leq L_{\max} < \infty$ then $m(\mathbf{y}) \leq L_{\max} \int \pi = L_{\max} < \infty$, and $m(\mathbf{y}) > 0$ because the integrand is positive on a set of positive measure. The logistic likelihood is a product of probabilities, so $L \leq 1$; the Poisson likelihood $\prod_i \mu_i^{y_i} e^{-\mu_i} / y_i!$, maximized freely over

$\boldsymbol{\mu} \in (0, \infty)^n$ at $\mu_i = y_i$, is bounded by $\prod_i y_i^{y_i} e^{-y_i} / y_i!$ (with $0^0 = 1$).

(b) Write $s_i = 2y_i - 1$ and use Eq. 35.3.1: $-\ell(\boldsymbol{\beta}) = \sum_i \log\{1 + \exp(-s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}$.

Overlap implies propriety. Since $\log(1 + e^{-t}) \geq \max(0, -t) = t_-$,

$$-\ell(\boldsymbol{\beta}) \geq \sum_{i=1}^n (s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})_- =: c(\boldsymbol{\beta}),$$

with c continuous and homogeneous of degree one. Under overlap $c(\mathbf{b}) > 0$ for every $\mathbf{b} \neq \mathbf{0}$, since $c(\mathbf{b}) = 0$ forces $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$ for all i , which overlap permits only at the origin. The unit sphere is compact, so $\delta = \min_{\|\mathbf{b}\|=1} c(\mathbf{b}) > 0$ and, by homogeneity, $c(\boldsymbol{\beta}) \geq \delta \|\boldsymbol{\beta}\|$. Hence $\exp\{\ell(\boldsymbol{\beta})\} \leq e^{-\delta \|\boldsymbol{\beta}\|}$, integrable over $\mathbb{R}^p$.

Separation implies impropriety. Let $\mathbf{b} \neq \mathbf{0}$ satisfy $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$ for all i , put $S = \{i : \mathbf{x}_{(i)}^\top \mathbf{b} = 0\}$, $\varrho = \min_{i \notin S} s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$, $\omega = \max_i \|\mathbf{x}_{(i)}\|$, and fix $\rho > 0$. For $\boldsymbol{\beta} = t\mathbf{b} + \mathbf{u}$ with $t \geq t_0 := 2\omega\rho/\varrho$ and $\|\mathbf{u}\| \leq \rho$, every $i \notin S$ has $s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} \geq t\varrho - \omega\rho \geq 0$ and every $i \in S$ has $s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} = s_i \mathbf{x}_{(i)}^\top \mathbf{u} \geq -\omega\rho$. So $-\ell(\boldsymbol{\beta}) \leq n \log\{1 + e^{\omega\rho}\}$ on that whole tube, giving $\exp\{\ell(\boldsymbol{\beta})\} \geq \{1 + e^{\omega\rho}\}^{-n}$ there; and the tube has infinite Lebesgue measure, containing a half-infinite cylinder about the ray. Hence $m(\mathbf{y}) = \infty$. ■

Warning. The second part is not a curiosity. Separation is common in small samples and with rare outcomes, and a “noninformative” Bayesian analysis of separated binary data summarizes a measure of infinite mass: whatever the sampler prints is an artefact of where it wandered. Example (The separated subgroup, three priors) makes the divergence visible.

39.4.2 The Laplace approximation

For large n the integral in Eq. 39.4.1 is dominated by a shrinking neighbourhood of the mode, where the integrand looks normal. Write $q_n(\boldsymbol{\beta}) = \ell(\boldsymbol{\beta}) + \log \pi(\boldsymbol{\beta})$ for the unnormalized log posterior, $\hat{\boldsymbol{\beta}}_n$ for its maximizer and $\mathbf{H}_n = -\nabla^2 q_n(\hat{\boldsymbol{\beta}}_n)$.

Theorem 39.4.2. Laplace approximation

Let $\rho_n = n^{-3/8}$ and $B_n = \{\boldsymbol{\beta} : \|\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_n\| \leq \rho_n\}$, and suppose there are constants $0 < c \leq C < \infty$ and $a, K > 0$, free of n , with:

- (L1) q_n is four times continuously differentiable on B_n , with $\nabla q_n(\hat{\boldsymbol{\beta}}_n) = \mathbf{0}$ and $\mathbf{H}_n$ positive definite;
- (L2) $cn \leq \lambda_{\min}(\mathbf{H}_n) \leq \lambda_{\max}(\mathbf{H}_n) \leq Cn$, and every third and fourth partial derivative of q_n is bounded in absolute value by Cn on B_n ;
- (L3) $\int_{\mathbb{R}^p \setminus B_n} \exp\{q_n(\boldsymbol{\beta}) - q_n(\hat{\boldsymbol{\beta}}_n)\} d\boldsymbol{\beta} \leq K \exp(-an^{1/4})$.

Then

$$\int_{\mathbb{R}^p} e^{q_n(\boldsymbol{\beta})} d\boldsymbol{\beta} = (2\pi)^{p/2} \det(\mathbf{H}_n)^{-1/2} e^{q_n(\hat{\boldsymbol{\beta}}_n)} \{1 + O(n^{-1})\}. \quad (39.4.3)$$

Proof. Write I_n for the integral and $J_n = I_n \det(\mathbf{H}_n)^{1/2} e^{-q_n(\hat{\boldsymbol{\beta}}_n)}$, so the claim is $J_n = (2\pi)^{p/2} \{1 + O(n^{-1})\}$. Substitute $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}_n + \mathbf{H}_n^{-1/2} \mathbf{u}$, Jacobian $\det(\mathbf{H}_n)^{-1/2}$:

$$J_n = \int_{\mathbb{R}^p} \exp\{q_n(\hat{\boldsymbol{\beta}}_n + \mathbf{H}_n^{-1/2} \mathbf{u}) - q_n(\hat{\boldsymbol{\beta}}_n)\} d\mathbf{u}.$$

Because $\nabla q_n(\hat{\boldsymbol{\beta}}_n) = \mathbf{0}$, Taylor's theorem on B_n gives

$$q_n(\hat{\boldsymbol{\beta}}_n + \mathbf{H}_n^{-1/2} \mathbf{u}) - q_n(\hat{\boldsymbol{\beta}}_n) = -\frac{1}{2} \|\mathbf{u}\|^2 + A_n(\mathbf{u}) + r_n(\mathbf{u}),$$

where A_n is the cubic form built from the third derivatives at $\hat{\boldsymbol{\beta}}_n$ and r_n is the fourth-order remainder. By (L2) the derivatives are bounded by Cn and $\|\mathbf{H}_n^{-1/2} \mathbf{u}\| \leq (cn)^{-1/2} \|\mathbf{u}\|$, so

$$|A_n(\mathbf{u})| \leq c_3 n^{-1/2} \|\mathbf{u}\|^3, \quad |r_n(\mathbf{u})| \leq c_4 n^{-1} \|\mathbf{u}\|^4, \quad (39.4.4)$$

with $c_3 = p^3 C / (6c^{3/2})$ and $c_4 = p^4 C / (24c^2)$, valid whenever $\hat{\boldsymbol{\beta}}_n + \mathbf{H}_n^{-1/2} \mathbf{u} \in B_n$.

Let $S_n = \{\mathbf{u} : \|\mathbf{u}\| \leq \sqrt{cn}^{1/8}\}$ and U_n the image of B_n . Every $\mathbf{u} \in S_n$ has $\|\mathbf{H}_n^{-1/2} \mathbf{u}\| \leq (cn)^{-1/2} \|\mathbf{u}\| \leq n^{-3/8} = \rho_n$, so $S_n \subseteq U_n$; and every $\mathbf{u} \in U_n$ has $\|\mathbf{u}\| \leq \lambda_{\max}(\mathbf{H}_n)^{1/2} \rho_n \leq \sqrt{C} n^{1/8}$. Split J_n over S_n , over $U_n \setminus S_n$ and outside U_n .

Outside U_n . Changing back to $\boldsymbol{\beta}$ and using (L3) with $\det(\mathbf{H}_n)^{1/2} \leq (Cn)^{p/2}$ bounds that piece by $(Cn)^{p/2} K e^{-an^{1/4}}$, which is $o(n^{-1})$.

Between S_n and U_n . There $\|\mathbf{u}\| \leq \sqrt{C} n^{1/8}$, so Eq. 39.4.4 gives $|A_n| + |r_n| \leq c_3 C^{3/2} n^{-1/8} + c_4 C^2 n^{-1/2} \leq 1$ for large n ; the integrand is then at most $e^{1 - \|\mathbf{u}\|^2/2}$, whose integral over $\|\mathbf{u}\| > \sqrt{c} n^{1/8}$ is $O(e^{-cn^{1/4}/4}) = o(n^{-1})$.

Inside S_n . Put $R_n = A_n + r_n$, so $|R_n| \leq 1$ there for large n . From $|e^x - 1 - x - \frac{1}{2}x^2| \leq \frac{1}{6}|x|^3 e^{|x|}$ for $|x| \leq 1$,

$$\begin{aligned} \int_{S_n} e^{-\|\mathbf{u}\|^2/2} e^{R_n} d\mathbf{u} &= \int_{S_n} e^{-\|\mathbf{u}\|^2/2} \left\{1 + R_n + \frac{1}{2} R_n^2\right\} d\mathbf{u} + \mathcal{E}_n, \\ |\mathcal{E}_n| &\leq \frac{\varepsilon}{6} \int_{S_n} e^{-\|\mathbf{u}\|^2/2} |R_n|^3 d\mathbf{u}. \end{aligned}$$

Four pieces. First, $\int_{S_n} e^{-\|\mathbf{u}\|^2/2} d\mathbf{u} = (2\pi)^{p/2} + O(e^{-cn^{1/4}/2})$. Second, A_n is homogeneous of degree three, hence odd, and S_n is a ball about the origin, so $\int_{S_n} e^{-\|\mathbf{u}\|^2/2} A_n(\mathbf{u}) d\mathbf{u} = 0$ exactly. Third, $\int_{S_n} e^{-\|\mathbf{u}\|^2/2} |r_n| d\mathbf{u} \leq c_4 n^{-1} \int e^{-\|\mathbf{u}\|^2/2} \|\mathbf{u}\|^4 d\mathbf{u} = O(n^{-1})$. Fourth, by Eq. 39.4.4 $R_n^2 \leq 2c_3^2 n^{-1} \|\mathbf{u}\|^6 + 2c_4^2 n^{-2} \|\mathbf{u}\|^8$, of Gaussian integral $O(n^{-1})$, and $|R_n|^3 = O(n^{-3/2} \|\mathbf{u}\|^9) + O(n^{-3} \|\mathbf{u}\|^{12})$, of Gaussian integral $O(n^{-3/2})$. Adding the three regions gives $J_n = (2\pi)^{p/2} + O(n^{-1})$, which is Eq. 39.4.3. ■

Remark (When the conditions hold). For a generalized linear model with a proper, positive, twice continuously differentiable prior and the conditions (G1)–(G4) of Theorem 34.5.1, (L1)

and (L2) hold with probability tending to one, because $\mathbf{H}_n = \mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X} / \phi + O(1)$ and (G4) makes its smallest eigenvalue grow linearly. Condition (L3) says the posterior concentrates; verifying it in general is a separate argument this book does not give. Kass and Raftery (1995, section 4) discuss the regularity conditions and the accuracy of the approximation.

Corollary 39.4.3. Where the BIC comes from

Under the conditions of [Theorem 39.4.2](#), suppose in addition that $\mathbf{H}_n/n \rightarrow \bar{\mathbf{H}}$ positive definite, that the prior is continuous and positive at the limit of $\hat{\boldsymbol{\beta}}_n$, and that $\hat{\boldsymbol{\beta}}$ maximizes ℓ with $\ell(\hat{\boldsymbol{\beta}}) - \ell(\hat{\boldsymbol{\beta}}_n) = O(n^{-1})$. Then

$$-2 \log m(\mathbf{y}) = -2\ell(\hat{\boldsymbol{\beta}}) + p \log n + O(1), \quad (39.4.5)$$

where the $O(1)$ term is $\log \det(\mathbf{H}_n/n) - p \log(2\pi) - 2 \log \pi(\hat{\boldsymbol{\beta}}_n) + o(1)$. The leading two terms are the BIC of Definition 29.2.2.

Proof. Take logarithms in [Eq. 39.4.3](#):

$$2 \log m(\mathbf{y}) = 2\ell(\hat{\boldsymbol{\beta}}_n) + 2 \log \pi(\hat{\boldsymbol{\beta}}_n) + p \log(2\pi) - \log \det \mathbf{H}_n + O(n^{-1}).$$

Write $\log \det \mathbf{H}_n = p \log n + \log \det(\mathbf{H}_n/n)$ and replace $\ell(\hat{\boldsymbol{\beta}}_n)$ by $\ell(\hat{\boldsymbol{\beta}})$ at a cost of $O(n^{-1})$; rearranging gives [Eq. 39.4.5](#) with the stated remainder, which converges because $\mathbf{H}_n/n \rightarrow \bar{\mathbf{H}}$. ■

The remainder in [Eq. 39.4.5](#) is $O(1)$, not $o(1)$: the BIC stays a bounded distance from $-2 \log m(\mathbf{y})$, and that distance carries the whole influence of the prior, so a BIC difference is only a crude approximation to a log Bayes factor ([Proposition 29.2.4](#)). What makes it useful is that a difference in dimension contributes $\log n$, which grows, while the neglected term does not. A third criterion belongs to the same family: the deviance information criterion of Spiegelhalter, Best, Carlin and van der Linde (2002) replaces p by an effective number of parameters estimated from posterior draws, which suits a model whose parameters are themselves shrunk by a prior.

Example. How good is the approximation

Take a logistic regression with an intercept and one covariate, a $N(0, 5^2)$ prior on each coefficient, and data simulated at $\boldsymbol{\beta} = (-0.4, 1.2)^\top$. Adaptive Gauss–Hermite quadrature on a tensor grid centred at the mode computes [Eq. 39.4.1](#) to machine accuracy, so the Laplace error can be measured exactly.

At $n = 25$ the Laplace marginal likelihood is too small by 0.10700, or eleven per cent; the relative error falls to 0.001114 at $n = 1600$. Regressing the log error on $\log n$ over seven sample sizes gives a slope of -1.06 , and over those seven sizes the product $n \times \text{error}$ stays between 1.530 and 2.675: the $O(n^{-1})$ of [Eq. 39.4.3](#), over a decade and a half of sample sizes ([Figure 39.4.1](#), panel (b)).

Panel (a) shows what that error looks like at $n = 25$: the marginal posterior of the slope is visibly right-skewed, mean 2.3291 against mode 1.9271, with total variation distance 0.1827

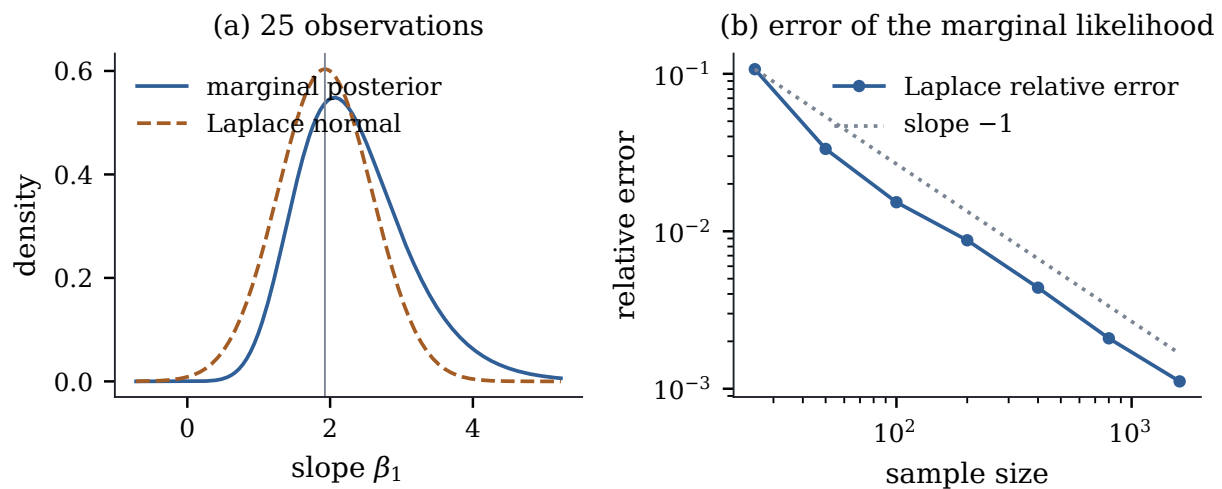


Figure 39.4.1: The Laplace approximation in a two-parameter logistic regression. (a) The marginal posterior of the slope at 25 observations against its normal approximation; the vertical line is the mode. (b) Relative error of the Laplace marginal likelihood against sample size, on log scales, with a reference line of slope -1 .

to the normal approximation. Reporting the mode with the curvature as a standard error would put the estimate in the wrong place and the interval on the wrong side. The corollary is visible too: over the same seven sizes the gap between the BIC and $-2 \log m(\mathbf{y})$ stays between -3.41 and -1.61 , an $O(1)$ that does not grow.

```
import numpy as np
from scipy import special, stats
TAU = 5.0                                     # prior standard deviation,  $N(0, \text{TAU})$ 
 $\sim I)$ 
BETA0 = np.array([-0.4, 1.2])                # the truth used to simulate

def simulate(n, seed):
    rng = np.random.default_rng(seed)
    X = np.column_stack([np.ones(n), rng.uniform(-2, 2, n)])
    y = rng.binomial(1, special.expit(X @ BETA0)).astype(float)
    return X, y

def log_post(beta, X, y):
    """Log likelihood plus log prior, up to a constant free of beta."""
    eta = X @ beta
    return np.sum(y * eta - np.logaddexp(0.0, eta)) - np.sum(beta**2) / (2 * TAU**2)

def mode_and_hessian(X, y):
    """Posterior mode and the negative Hessian there, by Newton's method."""
    beta = np.zeros(X.shape[1])
    for _ in range(100):
        pi = special.expit(X @ beta)
        H = X.T @ ((pi * (1 - pi))[:, None] * X) + np.eye(len(beta)) / TAU**2
        g = X.T @ (y - pi) - beta / TAU**2
        beta = beta + np.linalg.solve(H, g)
    return beta, H
```

```

def log_marginal_laplace(X, y):
    """log of the Laplace value of the integral of exp(log_post)."""
    beta, H = mode_and_hessian(X, y)
    p = len(beta)
    return (log_post(beta, X, y) + p / 2 * np.log(2 * np.pi)
            - 0.5 * np.linalg.slogdet(H)[1] - p / 2 * np.log(2 * np.pi * TAU**2))

def log_marginal_quadrature(X, y, nodes=60):
    """The same integral by adaptive Gauss-Hermite quadrature: the reference value.
    """
    beta, H = mode_and_hessian(X, y)
    p = len(beta)
    L = np.linalg.cholesky(np.linalg.inv(H))
    z, w = np.polynomial.hermite_e.hermegauss(nodes) # weight exp(-z^2/2)
    Z = np.stack(np.meshgrid(z, z, indexing="ij"), -1).reshape(-1, p)
    logw = np.log(w[:, None] + np.log(w)[None, :])
    pts = beta + Z @ L.T
    vals = np.array([log_post(b, X, y) for b in pts]) + (Z**2).sum(1) / 2 + logw.
    ravel()
    return (special.logsumexp(vals) + np.linalg.slogdet(L)[1]
            - p / 2 * np.log(2 * np.pi * TAU**2))

sizes = [25, 50, 100, 200, 400, 800, 1600]
rel_err = []
for k, n in enumerate(sizes):
    X, y = simulate(n, 3910 + k)
    a, b = log_marginal_laplace(X, y), log_marginal_quadrature(X, y)
    rel_err.append(abs(np.expm1(a - b)))
rel_err = np.array(rel_err)
slope = np.polyfit(np.log(sizes), np.log(rel_err), 1)[0]
print("n, relative error of the Laplace marginal likelihood:")
for n, e in zip(sizes, rel_err):
    print(f" {n:5d} {e:.3e} n * error = {n * e:.3f}")
print(f"log-log slope {slope:.3f}")

```

[Run this code online](#)

Two refinements are quoted without proof. For a positive smooth f , applying [Theorem 39.4.2](#) to $\exp\{q_n + \log f\}$ in the numerator of $E\{f(\boldsymbol{\beta}) \mid \mathbf{y}\} = \int f e^{q_n} / \int e^{q_n}$ raises the accuracy of the ratio to $O(n^{-2})$, the leading errors cancelling (Tierney and Kadane 1986); and the Bernstein–von Mises theorem says the posterior of $\mathbf{H}_n^{1/2}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_n)$ converges in total variation to $N_p(\mathbf{0}, \mathbf{I})$, so a credible set is asymptotically a confidence set (van der Vaart, 1998, chapter 10).

39.4.3 Sampling the posterior

When n is small, or the posterior skewed as in [Example \(How good is the approximation\)](#), an approximation at the mode is not enough. The alternative is a Markov chain whose stationary distribution is the posterior.

Definition 39.4.2. Metropolis–Hastings, Gibbs, and the two diagnostics

Let $\pi(\cdot \mid \mathbf{y})$ be a density known up to a constant.

- (a) The **Metropolis–Hastings** chain with proposal density $q(\cdot | \cdot)$ moves from $\boldsymbol{\beta}^{(t)}$ by drawing $\boldsymbol{\beta}^* \sim q(\cdot | \boldsymbol{\beta}^{(t)})$ and setting $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^*$ with probability

$$\alpha(\boldsymbol{\beta}^* | \boldsymbol{\beta}^{(t)}) = \min\left\{1, \frac{\pi(\boldsymbol{\beta}^* | \mathbf{y}) q(\boldsymbol{\beta}^{(t)} | \boldsymbol{\beta}^*)}{\pi(\boldsymbol{\beta}^{(t)} | \mathbf{y}) q(\boldsymbol{\beta}^* | \boldsymbol{\beta}^{(t)})}\right\}, \quad (39.4.6)$$

and $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)}$ otherwise. The normalizing constant cancels. A symmetric proposal, $q(\mathbf{b} | \mathbf{b}') = q(\mathbf{b}' | \mathbf{b})$, gives the **random-walk Metropolis** algorithm, in which α is the ratio of posterior densities capped at one.

- (b) The **Gibbs sampler** partitions $\boldsymbol{\psi} = (\boldsymbol{\psi}_1, \dots, \boldsymbol{\psi}_K)$ and replaces each block in turn by a draw from its **full conditional** $\pi(\boldsymbol{\psi}_k | \boldsymbol{\psi}_{-k}, \mathbf{y})$. It is the special case of (a) in which every proposal is accepted.
- (c) For c chains of length N , split each in half and let W be the mean of the $2c$ within-sequence variances and B the variance of the $2c$ sequence means. The **potential scale reduction** is

$$\hat{R} = \left\{ \frac{(N/2 - 1)W/(N/2) + B}{W} \right\}^{1/2}. \quad (39.4.7)$$

- (d) With ρ_t the lag- t autocorrelation of the stationary chain, the **effective sample size** of cN draws is $\text{ESS} = cN/(1 + 2 \sum_{t \geq 1} \rho_t)$, the sum truncated by Geyer's rule in the form used here: add the autocorrelations in consecutive pairs $\rho_{2k-1} + \rho_{2k}$, $k = 1, 2, \dots$, and stop at the first pair that is negative.

Proposition 39.4.4. The chain has the right stationary distribution

Let π be a density on $\mathbb{R}^p$ and $q(\cdot | \cdot)$ a transition density with $q(\mathbf{b}' | \mathbf{b}) > 0$ whenever $q(\mathbf{b} | \mathbf{b}') > 0$. The Metropolis–Hastings kernel of Eq. 39.4.6 satisfies detailed balance with respect to π :

$$\pi(\mathbf{b}) q(\mathbf{b}' | \mathbf{b}) \alpha(\mathbf{b}' | \mathbf{b}) = \pi(\mathbf{b}') q(\mathbf{b} | \mathbf{b}') \alpha(\mathbf{b} | \mathbf{b}') \quad \text{for all } \mathbf{b} \neq \mathbf{b}',$$

and consequently π is a stationary distribution of the chain.

Proof. Fix $\mathbf{b} \neq \mathbf{b}'$ and write $u = \pi(\mathbf{b})q(\mathbf{b}' | \mathbf{b})$, $v = \pi(\mathbf{b}')q(\mathbf{b} | \mathbf{b}')$. By Eq. 39.4.6, $\alpha(\mathbf{b}' | \mathbf{b}) = \min(1, v/u)$ and $\alpha(\mathbf{b} | \mathbf{b}') = \min(1, u/v)$, so both sides of the identity equal $\min(u, v)$. For stationarity let $P(\mathbf{b}, \cdot)$ be the kernel; for any measurable A ,

$$\int P(\mathbf{b}, A) \pi(\mathbf{b}) d\mathbf{b} = \int_A \int \pi(\mathbf{b}) q(\mathbf{b}' | \mathbf{b}) \alpha(\mathbf{b}' | \mathbf{b}) d\mathbf{b} d\mathbf{b}' + \int_A \pi(\mathbf{b}') \varrho(\mathbf{b}') d\mathbf{b}',$$

where $\varrho(\mathbf{b}') = 1 - \int q(\mathbf{b}'' | \mathbf{b}') \alpha(\mathbf{b}'' | \mathbf{b}') d\mathbf{b}''$ is the probability of staying put. By detailed balance the inner integral of the first term is $\pi(\mathbf{b}') \int q(\mathbf{b} | \mathbf{b}') \alpha(\mathbf{b} | \mathbf{b}') d\mathbf{b} = \pi(\mathbf{b}') \{1 - \varrho(\mathbf{b}')\}$, so the two terms add to $\int_A \pi(\mathbf{b}') d\mathbf{b}'$. ■

Detailed balance is all that is checked here; convergence to π from an arbitrary start needs irreducibility and aperiodicity, and a central limit theorem for the ergodic averages more still (Tierney 1994). The one design decision is the proposal. A covariance $\varsigma^2 \boldsymbol{\Sigma}_{\text{prop}}$ mixes well only when $\boldsymbol{\Sigma}_{\text{prop}}$ resembles the posterior covariance, and the Laplace matrix $\mathbf{H}_n^{-1}$ of Theorem 39.4.2

is the obvious candidate, at the cost of one Newton solve. Roberts, Gelman and Gilks (1997) showed that in a high-dimensional limit the optimal scale is $\varsigma = 2.38/\sqrt{p}$, with acceptance rate near 0.234 — derived for a product target, used everywhere as a rule of thumb.

39.4.4 Data augmentation for the probit model

A Gibbs sampler needs full conditionals that can be drawn from, and Proposition 39.3.1 says the conditional of $\boldsymbol{\beta}$ is not normal. Albert and Chib (1993) observed that enlarging the parameter space restores normality, through the latent variable representation of the probit model (Definition 35.2.1).

Proposition 39.4.5. The Albert–Chib sampler

Consider the ungrouped probit model $\Pr(Y_i = 1) = \Phi(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ with prior $\boldsymbol{\beta} \sim N_p(\mathbf{m}, \boldsymbol{\Sigma})$. Introduce latent variables $Z_i \mid \boldsymbol{\beta} \sim N(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}, 1)$, independent, and set $Y_i = \mathbf{1}\{Z_i > 0\}$. Then:

- (a) the marginal distribution of $\mathbf{Y}$ given $\boldsymbol{\beta}$ is the probit model, so the posterior of $\boldsymbol{\beta}$ is unchanged;
- (b) given $\mathbf{z}$ and $\mathbf{y}$, $\boldsymbol{\beta} \sim N_p(\mathbf{V}(\mathbf{X}^\top \mathbf{z} + \boldsymbol{\Sigma}^{-1} \mathbf{m}), \mathbf{V})$ with $\mathbf{V} = (\mathbf{X}^\top \mathbf{X} + \boldsymbol{\Sigma}^{-1})^{-1}$;
- (c) given $\boldsymbol{\beta}$ and $\mathbf{y}$, the Z_i are independent, Z_i being $N(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}, 1)$ truncated to $(0, \infty)$ if $y_i = 1$ and to $(-\infty, 0)$ if $y_i = 0$.

Proof.

- (a) $\Pr(Y_i = 1 \mid \boldsymbol{\beta}) = \Pr(Z_i > 0 \mid \boldsymbol{\beta}) = \Phi(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$, and the Z_i are independent given $\boldsymbol{\beta}$, hence so are the Y_i .
- (b) The joint density of $(\boldsymbol{\beta}, \mathbf{z}, \mathbf{y})$ is proportional to

$$\exp\left\{-\frac{1}{2}(\boldsymbol{\beta} - \mathbf{m})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\beta} - \mathbf{m}) - \frac{1}{2}\|\mathbf{z} - \mathbf{X}\boldsymbol{\beta}\|^2\right\} \prod_i \mathbf{1}\{\text{sign}(z_i) \text{ matches } y_i\}.$$

With $\mathbf{z}$ fixed the indicator is constant in $\boldsymbol{\beta}$, and the exponent is that of the conjugate normal update of Exercise (A2) with $\sigma^2 = 1$, response $\mathbf{z}$ and prior precision $\boldsymbol{\Sigma}^{-1}$; completing the square as in the proof of Theorem 7.6.2 gives the stated normal law.

- (c) With $\boldsymbol{\beta}$ fixed, the joint density factors over i into $\phi(z_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ times the indicator, which is the stated truncated normal density up to its normalizing constant. ■

The price is n extra draws per sweep; the gain is that every step is accepted and nothing needs tuning. The same device works for the logit link by a different route — the logistic distribution is a scale mixture of normals (Holmes and Held 2006), the binomial likelihood a mixture over Pólya–gamma variables (Polson, Scott and Windle, 2013) — and, more intricately, for multinomial and ordinal models.

Example. The 1996 survey, sampled

Return to the logistic regression of Example (Voting intention in the 1996 American election): the intended vote of 944 respondents on party identification, age in decades, education and

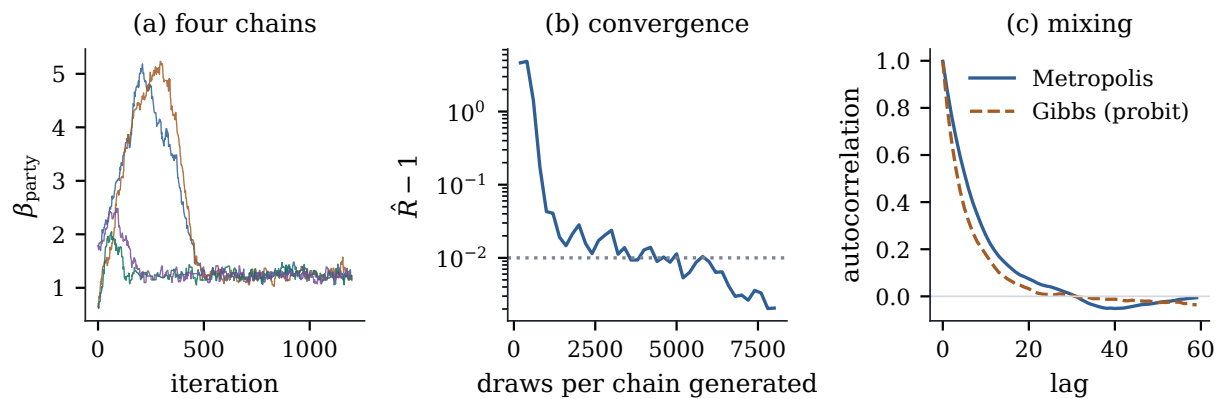


Figure 39.4.2: Diagnostics for the samplers of [Example \(The 1996 survey, sampled\)](#). (a) The first 1200 draws of four Metropolis chains for the party-identification coefficient, from starts eight posterior standard deviations away in every coordinate. (b) $\hat{R} - 1$, on a log scale, against draws generated per chain, recomputed from scratch at each length after discarding the first half; the reference line is $\hat{R} = 1.01$. (c) Autocorrelation functions of the Metropolis and Albert–Chib chains.

income. Put an independent $N(0, 10^2)$ prior on each of the 5 coefficients — nearly flat on this scale — and run four random-walk Metropolis chains of 8000 draws from starts 8 posterior standard deviations away in every coordinate, with the Laplace covariance as the proposal shape and scale $2.38/\sqrt{5}$. The acceptance rate is 0.298, and discarding the first half of each chain leaves 16000 draws.

Panel (a) of [Figure 39.4.2](#) shows the four chains for the party coefficient: two of them climb far above the mode before turning back, and all four are indistinguishable once the first few hundred iterations are past. Panel (b) applies the whole recipe — discard the first half, then split what is left — at each run length, and plots $\hat{R} - 1$ on a log scale. It starts at 5.63 with 200 draws per chain, is below 1.05 by 1000 and below 1.02 by 1400, and its own Monte Carlo variability keeps it brushing the 1.01 line until about 6000; at the full eight thousand the largest value over the five coefficients is 1.0066. Effective sample size is the binding constraint: 1079, or 0.067 effective draws per iteration.

The posterior mean of that coefficient is 1.2300 with posterior standard deviation 0.0705, against the maximum likelihood 1.2188 and standard error 0.0709: with $n = 944$ and a prior this weak the two analyses agree to the second decimal, as [Theorem 39.4.2](#) and the Bernstein–von Mises theorem say they must. What the simulation buys is the ability to ask for any posterior summary at all.

The probit version, sampled by [Proposition 39.4.5](#) with 4000 draws per chain, gives posterior mean 0.6780 against the maximum likelihood 0.6733; the ratio of the logistic coefficient to it is 1.814, against the $\pi/\sqrt{3} = 1.8138$ of [Proposition 35.2.2](#). Its effective sample size per iteration is 0.092, half again as good and with no tuning, though each iteration costs n truncated normal draws. Panel (c) compares the two autocorrelation functions.

```
import numpy as np
import statsmodels.api as sm
from scipy import special, stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["vote"].to_numpy(float) # 1 if the respondent
    expected to vote Dole
X = np.column_stack([np.ones(len(y)), anes["PID"], anes["age"] / 10.0,
```

```

        anes["educ"], anes["income"]])

n, p = X.shape
TAU = 10.0                                # prior: beta ~ N(0, TAU^-2
        I)

mle = sm.GLM(y, X, family=sm.families.Binomial()).fit()
w = mle.fittedvalues * (1 - mle.fittedvalues)    # the working weights at
        the mode
V_prop = np.linalg.inv(X.T @ (w[:, None] * X) + np.eye(p) / TAU**2)    # Laplace
        covariance
CHOL = np.linalg.cholesky(V_prop)
print("maximum likelihood:", mle.params.round(4))

```

[Run this code online](#)

```

def log_post(beta):
    eta = X @ beta
    return np.sum(y * eta - np.logaddexp(0.0, eta)) - np.sum(beta**2) / (2 * TAU**2)

def metropolis(n_draws, seed, start, step=2.38 / np.sqrt(5)):
    """Random-walk Metropolis with the Laplace covariance as the proposal shape."""
    rng = np.random.default_rng(seed)
    beta, lp = start.copy(), log_post(start)
    out, accepted = np.empty((n_draws, p)), 0
    for t in range(n_draws):
        prop = beta + step * (CHOL @ rng.standard_normal(p))
        lp_prop = log_post(prop)
        if np.log(rng.random()) < lp_prop - lp:    # the acceptance
            probability
            beta, lp, accepted = prop, lp_prop, accepted + 1
        out[t] = beta
    return out, accepted / n_draws

def split_rhat(draws):
    """Gelman-Rubin statistic on the two halves of every chain."""
    c, N = draws.shape
    s = draws.reshape(2 * c, N // 2)
    means, variances = s.mean(axis=1), s.var(axis=1, ddof=1)
    B, W = means.var(ddof=1), variances.mean()
    return np.sqrt(((N // 2 - 1) / (N // 2) * W + B) / W)

def ess(draws):
    """Effective sample size, with Geyer's initial positive sequence."""
    c, N = draws.shape
    centred = draws - draws.mean(axis=1, keepdims=True)
    var = draws.var(ddof=1)
    rho = np.array([np.mean([np.dot(x[:N - t], x[t:]) / N for x in centred]) / var
        for t in range(1, min(N, 400))])
    pairs = rho[: 2 * (len(rho) // 2)].reshape(-1, 2).sum(axis=1)
    k = np.argmax(pairs < 0) if np.any(pairs < 0) else len(pairs)
    return c * N / (1 + 2 * pairs[:k].sum())

short, short_rates = zip(*[metropolis(1500, 3960 + c, mle.params) for c in range(2)
    ])

```

```

short = np.array(short)[: , 750: , :]
print(f"two short chains: acceptance {np.mean(short_rates):.3f}, "
      f"party id mean {short[: , :, 1].mean():.4f}, R-hat {split_rhat(short[: , :, 1])
      :.4f}, "
      f"ESS {ess(short[: , :, 1]):.0f}")

```

[Run this code online](#)

```

def albert_chib(n_draws, seed, start):
    """Gibbs for the probit model: latent normals, then a normal draw for beta."""
    rng = np.random.default_rng(seed)
    V = np.linalg.inv(X.T @ X + np.eye(p) / TAU**2)
    L = np.linalg.cholesky(V)
    beta, out = start.copy(), np.empty((n_draws, p))
    for t in range(n_draws):
        eta = X @ beta
        lower = stats.norm.cdf(-eta) # P(latent < 0)
        u = np.where(y > 0, lower + (1 - lower) * rng.random(n), lower * rng.random(
n))
        z = eta + stats.norm.ppf(np.clip(u, 1e-12, 1 - 1e-12))
        mean = V @ (X.T @ z)
        beta = mean + L @ rng.standard_normal(p)
        out[t] = beta
    return out

gibbs_short = np.array([albert_chib(600, 3970 + c, np.zeros(p)) for c in range(2)])
print(f"short Gibbs run: probit party id mean {gibbs_short[: , :, 1].mean():.4f}, "
      f"R-hat {split_rhat(gibbs_short[: , :, 1]):.4f}")

```

[Run this code online](#)

39.4.5 Checking the fit from the posterior

Definition 39.4.3. Posterior predictive distribution and check

The **posterior predictive** distribution of a replicate data set $\mathbf{Y}^{\text{rep}}$ is

$$p(\mathbf{y}^{\text{rep}} | \mathbf{y}) = \int f(\mathbf{y}^{\text{rep}} | \boldsymbol{\beta}) \pi(\boldsymbol{\beta} | \mathbf{y}) d\boldsymbol{\beta}, \quad (39.4.8)$$

which is sampled by drawing $\boldsymbol{\beta}$ from the posterior and then $\mathbf{y}^{\text{rep}}$ from the model. For a statistic T the **posterior predictive p -value** is $\Pr\{T(\mathbf{Y}^{\text{rep}}) \geq T(\mathbf{y}) | \mathbf{y}\}$, or the lower-tail version $\Pr\{T(\mathbf{Y}^{\text{rep}}) \leq T(\mathbf{y}) | \mathbf{y}\}$ when small values of T are the departure of interest, estimated by the proportion of replicates on the relevant side of the observed value.

The check is only as good as the statistic. One the model was fitted to match — the total number of successes in a model with an intercept — has a p -value near 1/2 by construction; a useful statistic is one the model does *not* control, such as a subgroup total, a count of zeros or a maximum.

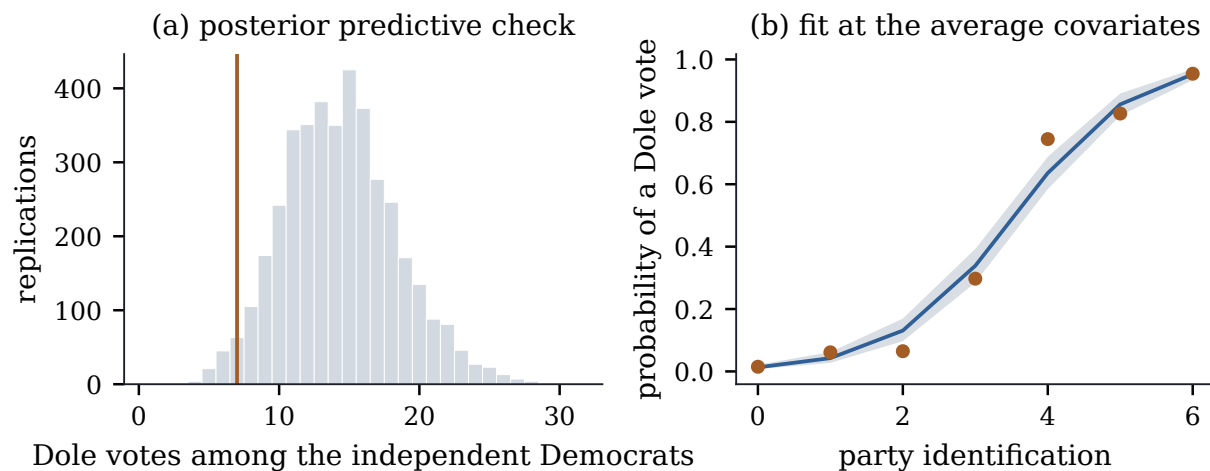


Figure 39.4.3: (a) Posterior predictive distribution of the number of Dole votes among the 108 independent Democrats, with the observed value marked. (b) Fitted probability of a Dole vote against party identification at the average of the other covariates, with a 95 per cent pointwise credible band and the raw category proportions (unadjusted).

Warning. The data are used twice: once to form the posterior and again to compute $T(\mathbf{y})$. The resulting p -value is conservative — its distribution under the model is concentrated towards $1/2$ rather than uniform — so a middling value is weak evidence of nothing while an extreme one is strong. Bayarri and Berger (2000) quantify the conservatism; Meng (1994) and Gelman, Meng and Stern (1996) give the general framework, including discrepancies $T(\mathbf{y}, \boldsymbol{\beta})$ that depend on the parameter too.

Example. Where the survey model fails

The model of [Example \(The 1996 survey, sampled\)](#) treats party identification as a single linear term. Take T to be the number of Dole votes among the 108 respondents who called themselves independent Democrats — a subgroup the model has no parameter for. Observed: 7; across four thousand posterior predictive replicates the mean is 14.4, and only 0.034 of them are as low as seven ([Figure 39.4.3](#), panel (a)). Given the conservatism just described, that is a real signal.

It is not one a quadratic term would catch: adding PID^2 drops the deviance by 0.019, while treating party identification as a seven-level factor drops it by 12.68 on five degrees of freedom, $p = 0.027$ — the same finding from the likelihood side. Panel (b) shows why: at average covariates the fitted curve passes well above the observed proportion at party identification two and below it at four. The monotone logistic curve is not flexible enough near the middle of this scale.

```
group = anes["PID"].to_numpy() == 2                                # the independent
Democrats
observed = y[group].sum()
short4 = np.array([metropolis(2000, 3980 + c, mle.params)[0] for c in range(4)][:,
                  1000:, :])
draws_short = short4.reshape(-1, p)
rng2 = np.random.default_rng(3990)
idx2 = rng2.choice(len(draws_short), 2000, replace=False)
```

```

rep_short = np.array([rng2.binomial(1, special.expit(X @ draws_short[i]))[group].sum()
                      for i in idx2])
print(f"short run: {int(observed)} of {int(group.sum())} against {rep_short.mean():.1f} "
      f"replicated; posterior predictive p-value {np.mean(rep_short <= observed):.3f} ")

```

[Run this code online](#)

39.4.6 Exercises

39.4.6.1 A. Check your understanding

Exercise. A1

Show that for a symmetric proposal the acceptance probability [Eq. 39.4.6](#) reduces to $\min\{1, \exp(q_n(\beta^*) - q_n(\beta^{(t)}))\}$, and that an uphill move is always accepted.

Solution. Symmetry gives $q(\beta^{(t)} | \beta^*) = q(\beta^* | \beta^{(t)})$, so those factors cancel and $\alpha = \min\{1, \pi(\beta^* | \mathbf{y})/\pi(\beta^{(t)} | \mathbf{y})\}$. The normalizing constant cancels too, leaving the ratio of the unnormalized densities, which is $\exp\{q_n(\beta^*) - q_n(\beta^{(t)})\}$. If $q_n(\beta^*) \geq q_n(\beta^{(t)})$ the ratio is at least one and $\alpha = 1$.

Exercise. A2

A chain is run once, from a start far from the posterior mode, and $\hat{R}$ computed from its two halves is 1.00. Is that evidence of convergence? What does the definition in [Eq. 39.4.7](#) require?

Solution. No. $\hat{R}$ near one says only that the pieces compared agree; a single chain stuck in one region has two halves that agree perfectly. The definition compares $2c$ sequences, and the value of the diagnostic comes from starting the c chains *over-dispersed relative to the posterior*, so that disagreement is detectable. A single chain also cannot reveal a second mode it never visited.

39.4.6.2 B. Practice

Exercise. B1

Take $q_n(\beta) = n g(\beta)$ for a fixed smooth g with a unique interior maximum at β^* and $g''(\beta^*) < 0$, and suppose g is bounded away from its maximum outside any neighbourhood of β^* . Verify (L1)–(L3) of [Theorem 39.4.2](#) and write out the approximation explicitly.

Exercise. B2

[Corollary 39.4.3](#) says the prior enters only the $O(1)$ remainder. Show that if the prior is

itself allowed to depend on n — say $\boldsymbol{\beta} \sim N_p(\mathbf{0}, n\boldsymbol{\Sigma}_0)$ with $\boldsymbol{\Sigma}_0$ fixed, so that the prior is made vaguer as data accumulate — then the remainder grows like $p \log n$, and say what that does to the BIC.

Solution. With π the $N_p(\mathbf{0}, n\boldsymbol{\Sigma}_0)$ density, $-2 \log \pi(\hat{\boldsymbol{\beta}}_n) = p \log(2\pi) + p \log n + \log \det \boldsymbol{\Sigma}_0 + \hat{\boldsymbol{\beta}}_n^\top \boldsymbol{\Sigma}_0^{-1} \hat{\boldsymbol{\beta}}_n / n$, so the term $-2 \log \pi(\hat{\boldsymbol{\beta}}_n)$ in the remainder grows like $p \log n$ instead of staying bounded. The expansion then reads $-2 \log m(\mathbf{y}) = -2\ell(\hat{\boldsymbol{\beta}}) + 2p \log n + O(1)$: a prior whose spread grows with n doubles the dimension penalty. This is Lindley’s paradox in miniature — a prior that is made vaguer as data accumulate penalizes extra parameters more and more heavily — and it is why the BIC’s derivation assumes a fixed prior.

Exercise. B3

A chain of 16 000 draws has effective sample size 1000. A colleague thins it by keeping every sixteenth draw, obtaining 1000 nearly independent draws, and says nothing has been lost. Is that right? What is thinning good for?

39.4.6.3 C. Going deeper

Exercise. C1

Show that the Gibbs update of a block $\boldsymbol{\psi}_k$ is a Metropolis–Hastings step with proposal $q(\boldsymbol{\psi}^* | \boldsymbol{\psi}) = \pi(\boldsymbol{\psi}_k^* | \boldsymbol{\psi}_{-k}, \mathbf{y})$ and that its acceptance probability is one. Deduce that [Proposition 39.4.4](#) covers the Gibbs sampler, and explain why a Gibbs sampler can nevertheless fail to be irreducible.